When Does Relational Structure Help Healthcare Fraud Detection? A Controlled Benchmark of Heterogeneous Graph Neural Networks and Graph-Augmented Tabular Models

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Abstract

Purpose: Graph neural networks (GNNs) are proposed for healthcare insurance fraud because collusive provider rings leave traces in claim networks, but when they beat tabular models is unclear. **Methods:** We built a synthetic claims benchmark that independently varies the feature-level signal and doctor–patient density of fraud rings, tests on rings of unseen members, and builds features per audit period to avoid leakage. Fourteen equally tuned methods, including gradient boosting with and without engineered graph features and five heterogeneous GNNs, were compared in nine regimes (six seeds each), a low-ring-volume stress test, and public provider-labelled Medicare claims (5,410 providers). **Results:** At default volume, where ring patients filed 5.7 times as many claims as others, per-entity aggregates detected most rings (average precision 0.89–0.93), and graph features added up to 0.10 for dense rings. At low volume, aggregates were near chance (0.05), graph features reached 0.41–0.60 for moderately and densely linked rings, and pure GNNs at most 0.16. No pure GNN significantly outperformed gradient boosting with graph features in any setting; GNNs were 0.08–0.15 worse on isolated fraud, and routing to a GNN gained little. Recurring ring members inflated performance by up to 0.60. On the public data, gradient boosting on provider aggregates had the highest average precision (0.704).

Conclusion: Relational structure helps when collusion forms dense blocks, most when ring members are not otherwise unusual, and engineered shared-patient features captured it better than message passing alone. Graph models should be evaluated against graph-augmented tabular baselines on unseen entities.

Keywords: Healthcare fraud detection, Provider rings, Heterogeneous graph neural networks, Gradient boosting, Synthetic benchmark, Data leakage

1 Introduction

Health insurance fraud causes losses estimated in the tens of billions of dollars each year in the United States alone [1]. Detecting it is a routine task of payers' claims-processing and audit systems, which must decide from administrative claims data which providers and claims a limited team of investigators examines. Fraud takes two broad forms. In *isolated* fraud a single provider inflates or fabricates individual claims, so the evidence lies in the claim itself: amounts, procedure counts or coding that are unusual for the provider's specialty. In *collusive* fraud a group of providers and recruited patients file claims together. Each claim can look ordinary; the evidence lies in who bills whom—patients who see many members of the same small group of providers, and providers who share an unusually large fraction of their patients.

The second form motivates graph neural networks (GNNs) over the patient–provider–facility network, in particular heterogeneous architectures such as RGCN [2], HAN [3], HGT [4] and Simple-HGN [5]; GNNs of this kind have also been built specifically for health-insurance claims [6–8]. Whether GNNs are needed, however, is an empirical question. Gradient-boosted trees remain strong on tabular data [9], relational signals can be engineered as features, and benchmark studies outside healthcare report that tree ensembles with neighbourhood features often match or beat GNNs on graph anomaly detection [10, 11]. On public Medicare claims with provider fraud labels, machine-learning models given graph-centrality features outperformed GNNs at a fraction of the training cost [12]. A single dataset of this kind, however, cannot show *when* relational structure helps or *why*: its labels mark providers, not collusive groups, and neither the typology nor the strength of the relational signal can be varied.

Answering the question for healthcare claims is difficult for two reasons. First, public claims data with labelled collusion are not available, and real labels conflate fraud typologies; on the most widely used public provider-labelled dataset, reported results vary widely between studies and are rarely accompanied by significance tests or code [13]. Second, evaluation is fragile: identifiers that encode the label, statistics computed on the full dataset, and providers that appear in both training and test data inflate apparent performance [14, 15]. Fraud rings make the last problem acute, because a model can score a test claim correctly by recognising a provider it saw labelled during training instead of recognising coordination.

We therefore study the question in a controlled setting and check it on public provider-labelled data. Our contributions are:

1. **A regime-controllable benchmark.** A synthetic claims generator in which the feature-level signal of ring claims (δ) and the density of the ring's doctor–patient block (s) are varied independently, with an *emergent* protocol in which every test ring is formed by doctors and patients never seen in a training ring, and an audit-period design that keeps features and graphs free of future information (Section 3). Ring volume can be lowered so that ring members are not outliers in claim volume. Code, generator and tests are publicly available.
2. **A controlled comparison** of 14 methods—tabular models with and without engineered graph features, a multilayer perceptron (MLP) control, five heterogeneous GNNs, two hybrids and an unsupervised overlap score—with an equal tuning budget, nine regimes with six generator seeds each, a low-volume stress test, and paired statistics that treat the seed as the unit of replication, together with an evaluation of two-stage routing from gradient boosting to a GNN (Sections 4 and 6).
3. **Mechanism and memorisation analyses**: degree-preserving rewiring, single-factor ablations of HGT (depth, attention heads, relations, node inputs), an analysis of which inputs the trained models rely on, and a comparison of new versus recurring rings that quantifies how much apparent ring detection comes from recognising known members (Section 7). A re-analysis of a small fixed benchmark shows how such data can suggest the opposite conclusion (Section 9.3).
4. **A check on public data**: a provider-labelled Medicare claims dataset evaluated with a provider-disjoint protocol, equal tuning budgets and tests corrected for overlapping training sets (Section 5). It replicates, under stricter conditions, an earlier finding on public Medicare provider data [12]. Its labels identify providers, not rings, so it tests whether graph information helps provider-level detection, not whether rings are detected.

Our main findings are as follows. Claim features alone cannot reveal rings whose claims look legitimate. When ring members are unusually active, as in our default setting, per-entity aggregates capture most of the evidence; when they are not, aggregates fail and coordination structure carries almost all of it. Structure adds measurable value only when the ring's doctor–patient block is moderately or highly dense. Where it does, gradient boosting with a few engineered shared-patient features was as good as or better than every pure GNN we tested, and far better when ring volume was low; GNNs that also receive those features gained little in average precision (AP; +0.008 averaged over regimes at default volume, not statistically significant), and significantly only for sparse rings at low volume. GNNs are clearly worse on isolated fraud, because message passing dilutes claim-level anomalies. At default volume HGT's ring detection is structural: it collapses without degree encodings and survives with degree encodings as its only inputs. Routing high-risk claims from gradient boosting to a GNN gained nothing; fusing the two models' rankings gained at most 0.006 AP, as an ensemble rather than as a router. Evaluations that let test rings share members with training rings overstate ring detection, by up to 0.60 AP for a model that only sees entity attributes. On the public provider-labelled data, where labels show no relational assortativity, no graph-based method improved on gradient boosting over provider aggregates, in line with earlier work on these data [12].

2 Related Work

Tabular healthcare fraud detection.

Most published healthcare-fraud detectors are supervised classifiers over claim- or provider-level features. On Medicare data labelled with the federal exclusion list (LEIE), Bauder and Khoshgoftaar [16] compared classical learners, Herland et al. [17] combined several Medicare data sources, and Johnson and Khoshgoftaar [18] studied neural networks under extreme class imbalance. Earlier work on Medicaid data used multidimensional data models and outlier analysis [19]. These models score each claim or provider from its own attributes and pre-computed aggregates; relational information enters, if at all, through hand-engineered features. A recent review of classifiers for healthcare fraud found that results reported on the public provider-labelled dataset used in Section 5 range from an F1 score of 0.49 to 0.90 under differing protocols, and that significance tests and released code are rare [13].

Graph-based fraud and anomaly detection.

Surveys of graph-based anomaly detection [20, 21] document how coordination leaves structural traces such as dense blocks, lockstep behaviour and unusual egonets. Unsupervised detectors exploit these traces directly, e.g. OddBall [22], CopyCatch [23] and FRAUDAR [24]. GNNs have been applied to review fraud [25], underground-forum key players [26], and camouflaged or class-imbalanced fraudsters [27, 28]; spectral analyses argue that anomalies shift graph spectra in ways that standard message passing can blur [29].

Graph learning for health-insurance fraud.

Heterogeneous GNNs have been applied to health-insurance data from single payers: attention over meta-paths of patients, doctors, departments and medicines to classify patients [6], graph structure learning [7], and heterogeneous GNNs, among them a graph transformer, that classify individual billed activities, explained with GNNExplainer and PGExplainer [8]. These studies compare their models with other graph models [6] or with tabular models that receive no relational features [8]. On the public provider-labelled dataset we also use, a graph transformer network reached an F1 score of 0.53 against 0.52 for a random forest in a study of graph attention networks [30], as summarised in [13]; graph embeddings of meta-path graphs computed on a 10% claim sample reached an F1 score of 0.56 with an ROC-AUC similar to that of tree ensembles (0.87–0.88) [31]; and machine-learning models given graph-centrality features outperformed GNNs in precision, recall and F1 while training 250–300 times faster [12]. Our analysis of these data replicates the last finding under a stricter protocol. What these studies cannot show, because their labels mark providers, patients or activities rather than collusive groups, is under which conditions relational structure carries information that per-entity features miss.

Heterogeneous GNNs.

Relational GCN [2] uses relation-specific weights; HAN [3] aggregates over hand-chosen meta-paths with node- and semantic-level attention; HGT [4] parameterises

attention by node and edge type without meta-paths; and Simple-HGN [5] showed that a GAT with edge-type embeddings, residual attention and output normalisation is competitive with more elaborate heterogeneous architectures when baselines are tuned fairly.

When do graph models beat tabular models?

Evidence on this question is mixed. On the Elliptic Bitcoin dataset a random forest on node features outperformed graph convolutional networks under a temporal split [10]. GADBench [11] found that tree ensembles with simple neighbourhood aggregation are among the strongest supervised graph anomaly detectors across ten datasets, GraphLand found gradient-boosted trees with graph-based input features to be very strong, though not universally winning, baselines on industrial graphs [32], a graph gradient-boosting machine was on par with or better than the state of the art for insurance fraud [33], and tree ensembles remain hard to beat on tabular data generally [9]. Synthetic benchmarks with injected typologies make the question testable: Altman et al. [34] generate money-laundering patterns with known ground truth, and Jesus et al. [35] release controlled tabular fraud suites. We found no comparable controlled study for heterogeneous patient–provider–facility claim graphs, where the relevant typology is the collusive provider ring.

Evaluation pitfalls.

Leakage through features that encode the target, through preprocessing fitted on the full dataset, or through entities shared between training and test data inflates reported performance [14, 15]. GNN benchmarks are additionally sensitive to split choice and tuning effort [36], and duplicate nodes that leak between training and test data strongly affected GNN performance on two popular benchmarks [37]. These risks are acute for fraud rings, whose members recur across claims: a model can score a test claim correctly by recognising a provider it saw labelled in training rather than by recognising coordination.

3 Benchmark Design

3.1 Synthetic claims world

We simulate 6,000 patients, 300 doctors and 40 hospitals in five regions over four audit periods of six months (Fig. 1; Table A1). Each doctor has one of eight specialties, which sets the distribution of claim amounts, inpatient probability and procedure counts. Patients have heterogeneous utilisation (gamma-distributed claim rates with mean three claims per two years) and one to three regular doctors in their region; 85% of legitimate claims go to a regular doctor and the rest to a popularity-weighted random doctor. Claim amounts are log-normal given the specialty, procedure count and inpatient status.

Isolated fraud.

Three per cent of claims are upcoded claims filed through a patient's own regular doctor. Their log-amount, procedure count, inpatient probability and procedure-code

level are shifted by two standard deviations; their structure is that of an ordinary claim.

Provider rings.

A ring consists of four colluding doctors and twelve recruited patients, drawn preferentially from high utilisers. During its active period the ring files 16 claims per month between linked doctor–patient pairs, clustered around monthly burst days. Two parameters are varied independently:

- the *feature shift* $\delta \in \{0, 1, 2\}$ applied to ring claims, in standard deviations of log-amount (with matching shifts in procedures, inpatient probability and coding level). At $\delta = 0$ a ring claim is drawn from exactly the distribution of a legitimate claim of the same doctor's specialty;
- the *structural density* $s \in \{0.1, 0.5, 0.9\}$, the probability that a ring patient is linked to each ring doctor. Ring claim volume is fixed, so s changes only how the claims are spread over the doctor–patient block (realised block densities ≈ 0.37, 0.62 and 0.79).

Ring patients keep a single regular doctor, so the number of distinct doctors they see is governed by s. Their elevated claim volume, a degree signal visible to every model, is the same in all regimes and large: in a test period a ring patient files on average 9.3 claims, against 1.6 for other patients with a claim (99th percentile about 5), and a ring doctor 38 claims against 16. Section 6.6 reduces ring volume to four claims per month, which brings ring patients to 3.7 claims, about the 95th percentile of other patients, and ring doctors to 20.

Protocols.

Under the *emergent* protocol (default), four new rings start in each period and rings in different periods are formed by disjoint doctors and patients, so every test ring is unseen during training. Under the *persistent* protocol the same four rings are active in all periods, which reproduces the setting of most prior synthetic evaluations and lets a model succeed by recognising known members. Both protocols have identical claim and ring-claim volumes per period. Ground truth (fraud type and ring identifier) is stored separately from the claim records and never enters feature construction.

Replication.

Each regime is generated from six seeds. The generator draws a world's entities, legitimate claims, isolated fraud and recruited ring members before it applies the ring parameters, so for a given seed the nine regimes share all of these and differ only in their ring claims. Regimes are therefore compared on otherwise identical worlds, and analyses pooled over regimes treat the seed, not the regime–seed world, as the unit of replication (Section 4.5).

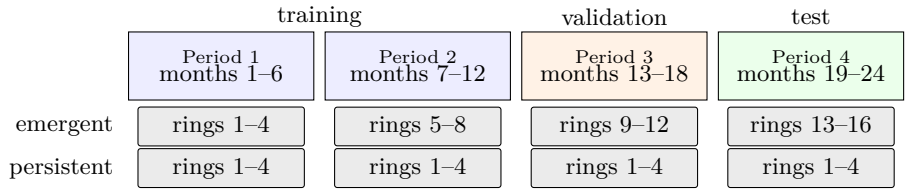


Fig. 1 Audit-period protocol. Each six-month period is analysed separately: features and graphs use only that period's claims. Under the emergent protocol every period has new rings formed by disjoint doctors and patients, so test rings are unseen; under the persistent protocol the same rings recur and can be recognised by their members.

3.2 Evaluation protocol

Periods 1–2 are used for training, period 3 for validation (early stopping, threshold selection) and period 4 for testing. Each period is analysed on its own, as in a retrospective audit: a claim's aggregate and graph features, and the graph a GNN operates on, are built only from claims of the same period. This keeps every split's features on the same time scale and prevents information from later periods reaching earlier ones; it is verified by automated tests (Section 10).

4 Methods

4.1 Tabular features

- **C** (claim): log-amount, procedures, length of stay, inpatient flag, coding level, month (cyclical), patient age, sex and chronic-condition count, doctor experience and specialty, hospital size.
- **A** (aggregates within the period, no labels): the patient's claim count, distinct doctors and mean/maximum log-amount; the doctor's claim count, distinct patients and mean log-amount; hospital claim count; patient–doctor pair count; and the claim's log-amount relative to the patient, doctor and specialty means.
- **G** (graph features on the period's patient–doctor bipartite graph, no labels): the doctor's maximum shared-patient Jaccard similarity with another doctor and number of doctors sharing at least two patients, the mean degree of the doctor's patients, the patient's mean doctor overlap and share of doctor pairs that co-treat other patients, and the triadic closure of the claim (how many of the patient's other doctors share patients with the claim's doctor), raw and degree-normalised.

4.2 Heterogeneous graph

Each period's graph has patient, doctor, hospital and claim nodes and the relations *files* (patient→claim), *billed_by* (claim→doctor), *affiliated* (doctor→hospital) and *visits* (patient→doctor), each with a reverse relation so that every node type receives messages. Claim nodes carry claim-intrinsic attributes; entity nodes carry their static attributes and $\log(1+\text{degree})$ for each incident relation, a label-free structural encoding included because attention- and mean-based aggregators cannot otherwise count neighbours [38]. No engineered aggregates or labels are used as node features (Fig. 2).

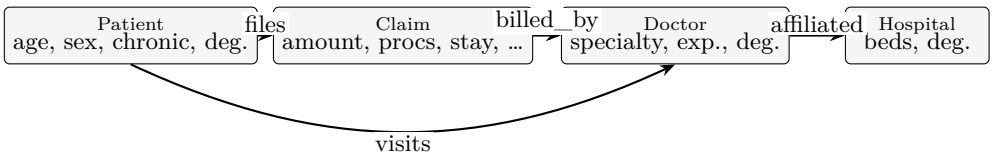


Fig. 2 Heterogeneous claim graph of one audit period. Every relation also has a reverse relation (not drawn), so every node type receives messages and doctor context reaches claim nodes. Entity nodes carry static attributes and log(1 + degree) per incident relation; claim nodes carry only claim-intrinsic attributes.

Claim attributes and degree encodings are standardised with statistics of the training periods; static entity attributes, which do not change over time and carry no labels, with statistics over all entities.

4.3 Models

- **Tabular**: logistic regression (LR) and random forest (RF) on C+A; histogram gradient boosting (GB) [39] on C, C+A and C+A+G.
- **MLP-C**: a multilayer perceptron on C, the no-graph control for the GNNs.
- **GNNs**: heterogeneous GraphSAGE [40] (one convolution per relation, mean or sum aggregation), RGCN [2], HAN [3] (claim nodes; meta-paths claim–patient–claim and claim–doctor–claim, with patient, doctor and hospital attributes appended to claim inputs), Simple-HGN [5] and HGT [4]. All share the same per-type input projection, residual connections with layer normalisation, and two-layer classification head.
- **Hybrids**: SAGE+AG and HGT+AG, whose claim nodes also receive the A and G features, testing whether message passing adds information beyond the engineered features. Hybrids reuse the tuned configuration of their base architecture and receive no separate tuning budget.
- **Overlap score**: the degree-normalised triadic closure of the claim, used directly as an unsupervised score.

Tabular models use scikit-learn [41] and graph models PyTorch Geometric [42]. Neural models are trained full-batch with AdamW, class-weighted binary cross-entropy and early stopping on validation average precision (patience 20, at most 200 epochs). Every model is trained once per world with the same training seed, so variation between worlds includes variation due to initialisation.

Hyperparameters.

Every method receives the same budget of up to ten configurations (the first being a default), selected by mean validation average precision on two development worlds (generator seeds 901 and 902, $\delta = 0.5$, $s = 0.5$) that are never used for evaluation (Appendix A, Table A2). The development regime ($\delta = 0.5$) is deliberately not one of the evaluation regimes.

4.4 Two-stage routing

A common deployment proposal is to score every claim with a cheap tabular model and pass only high-risk claims to a GNN. We evaluate this with GB-CAG as the first stage and HGT or SAGE as the second. A claim is escalated if its GB-CAG score is at least the $(1 - e)$ quantile of the GB-CAG scores in the validation period. The escalation fraction $e \in \{0.05, 0.1, 0.2, 0.3, 0.5, 1\}$ is chosen by validation AP, so test claims never influence it. Escalated claims rank above all other claims and are ordered either by the GNN score alone (*re-rank*: GB-CAG acts as a filter) or by the mean of the two models' validation percentiles (*fused*); non-escalated claims keep their GB-CAG order. With $e = 1$, re-ranking reduces to the GNN alone and fusion to a rank-average ensemble of the two models, so both single models and the ensemble are among the candidates. A claim's routed score depends only on its own two scores and on validation statistics. The stages are the models of the main experiment, retrained with the same seeds; their test predictions reproduce the stored ones exactly for the GNNs and to within 3×10^{-8} for GB-CAG (float32 storage).

4.5 Metrics and statistics

The primary metric is average precision (AP, the area under the precision–recall curve), appropriate under class imbalance [43]. *Ring AP* scores ring claims against legitimate claims (isolated fraud excluded); *isolated AP* is defined analogously. We also report F1 at the threshold that maximises validation F1, the area under the receiver operating characteristic curve (ROC-AUC) and, because investigators can examine only a limited number of cases [44], recall when the top 5% of claims are investigated. For routing we add precision among the top 5% of claims (P@5%) and, as a ring-level measure, the share of test rings with at least one claim in that budget.

Each of the nine regimes is replicated with six worlds (generator seeds 1–6), 54 worlds in total. Within a regime the six worlds are independent, and methods are compared with paired t-tests over them. Because the nine regimes of a seed share most of their data (Section 3.1), comparisons pooled over regimes average each seed's nine paired differences and test the six seed means (t-test with five degrees of freedom); treating the 54 worlds as independent would overstate the evidence. p-values are Holm-adjusted within each pre-specified family of comparisons [45], Wilcoxon signed-rank tests are a robustness check [46], and intervals are 95% confidence intervals (CIs) from the t-distribution over worlds or, for pooled comparisons, over seeds. A non-significant difference is not evidence that two methods are equivalent; the CIs show which differences remain compatible with the data.

4.6 Use of AI tools

An AI coding assistant (Claude, Anthropic) was used to help write and review the analysis code, run and document the experiments, and draft and edit parts of the text. All reported numbers are produced by the released code (Section 10). The authors reviewed all code, analyses and text and take full responsibility for the content.

5 Public Provider-Labelled Data

5.1 Data

We use the public "Healthcare Provider Fraud Detection Analysis" dataset [47] (CC0 public domain), which its publisher describes as Medicare inpatient and outpatient claims linked to beneficiaries, with each provider labelled as potentially fraudulent or not; graph models studied on these data are discussed in Section 2. Only the labelled training files are used: 558,211 claims (40,474 inpatient, 517,737 outpatient) dated between November 2008 and December 2009, filed by 5,410 providers (506, or 9.4%, labelled fraudulent) for 138,556 beneficiaries, with 100,737 distinct attending, operating or other physicians. The documentation describes neither the source of the claims nor how the labels were obtained, so we treat them as given and discuss this limitation in Section 9.6. Because labels exist only per provider, the prediction unit is the provider; claim-level ring detection cannot be evaluated on these data.

5.2 Features and graph

P (23 provider aggregates): claim volume and inpatient share, beneficiaries and claims per beneficiary, distinct attending and operating physicians, reimbursement totals and log-amount statistics, deductibles, claim duration and length of stay, diagnosis and procedure counts, share of claims with an operating physician, and the age, chronic-condition burden, mortality, renal-disease share, annual reimbursements and geographic spread of the provider's beneficiaries. **G** (8 label-free graph features): for both shared beneficiaries and shared physicians, the number of co-providers, the maximum Jaccard overlap with another provider, the share of the provider's beneficiaries (physicians) seen by other providers, and their mean number of providers. **L** (4 label-neighbourhood features): the share of co-providers that are known fraudulent *training* providers, and an indicator of having any labelled co-provider, through beneficiaries and through physicians. L uses only training labels and never a provider's own label.

The heterogeneous graph has provider, beneficiary and physician nodes, with relations *serves* (provider–beneficiary, 363,300 edges) and *employs* (provider–physician) plus reverse relations. Physicians working for a single provider carry no cross-provider information and are summarised in P rather than kept as nodes, leaving 7,685 shared physicians (16,287 edges). Provider nodes carry P (or P+G for the hybrids), beneficiary nodes 22 demographic and clinical attributes, and physician nodes their claim and provider counts. HAN uses the provider-only meta-paths provider–beneficiary–provider and provider–physician–provider.

5.3 Protocol

All providers' claims are observed and define the graph; labels of validation and test providers are hidden. This matches an audit in which claims are available for every provider but only some providers have been investigated. Providers are assigned to five stratified folds; in rotation k, fold k is the test set, fold $k+1$ the validation set and the remaining three folds the training set, so every provider is tested exactly once. Hyperparameters are selected once, with the same budget of six configurations

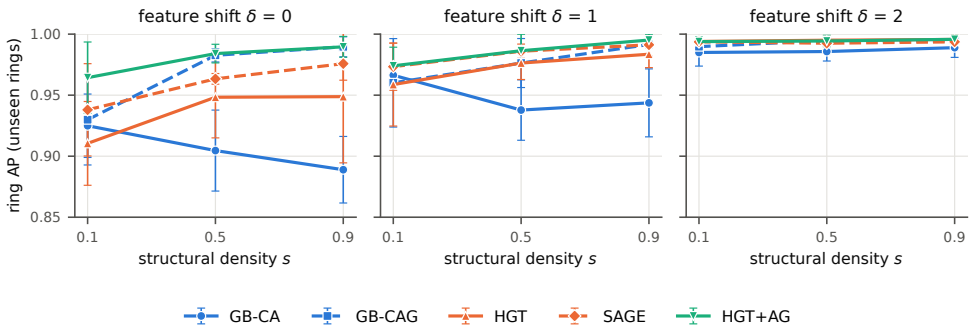


Fig. 3 Ring AP on unseen test rings by feature shift δ and structural density s (mean and 95% t-interval over six worlds). GB-CA: gradient boosting on claim and aggregate features; GB-CAG: plus engineered graph features; HGT+AG: HGT whose claim nodes also receive the A and G features. Feature-only models (GB-C, MLP-C) lie far below the plotted range (Table 1).

per method, on a separate stratified 60/20/20 split (seed 1000), and then fixed for the five rotations. Graph models are trained full-batch for at most 100 epochs with early stopping (patience 10) on validation AP. Metrics are AP, ROC-AUC, F1 at the validation-selected threshold, and recall among the top 10% of providers by score (the label prevalence is 9.4%). Because the rotations' training sets overlap, their results are positively correlated and a plain paired t-test over them is anti-conservative. Differences between methods are therefore tested with the corrected resampled t-test [48], which replaces the factor $1/J$ in the variance of the mean difference by $1/J + n_2/n_1$, where $J = 5$ is the number of rotations and $n_2/n_1 = 1/3$ the ratio of test to training providers; p-values are Holm-adjusted over all comparisons. With five rotations these tests have little power, so we also report a provider-level bootstrap of the difference in AP from the best method (1,000 resamples within each rotation's test fold and label, trained models held fixed); it reflects which providers happen to be tested, not retraining variability.

6 Results on the Controlled Benchmark

Each test period contains on average 4,924 claims, of which 384 belong to four unseen rings and 133 are isolated fraud, so the chance levels of ring AP and isolated AP are 0.080 and 0.029. Table 1 reports ring AP for every method and regime, Fig. 3 shows the main methods with 95% intervals, and Fig. 4 summarises paired differences. Table 2 averages the paired differences over the nine regimes, with the seed as the unit. Recall when the top 5% of claims are investigated cannot exceed 0.476, because fraud makes up 10.5% of test claims, and every method with aggregate features or a graph reached 0.43–0.48 in every regime; at this budget the methods differ little (supplementary results), so we compare them by AP, which covers the whole ranking.

Table 1 Ring-fraud detection on unseen test rings: average precision of ring versus legitimate claims (chance level 0.08), mean over six worlds per regime. Bold: best per regime. Standard deviations are in the supplementary results files.

	$\delta = 0$			$\delta = 1$			$\delta = 2$		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.084	0.086	0.087	0.329	0.326	0.334	0.805	0.818	0.809
MLP-C	0.088	0.087	0.095	0.353	0.358	0.360	0.803	0.817	0.808
LR-CA	0.923	0.888	0.868	0.968	0.958	0.952	0.988	0.986	0.987
RF-CA	0.875	0.838	0.844	0.957	0.924	0.941	0.985	0.983	0.986
GB-CA	0.925	0.905	0.889	0.966	0.938	0.944	0.985	0.986	0.989
GB-CAG	0.930	0.983	**0.990**	0.960	0.976	0.992	0.990	0.995	0.994
SAGE	0.938	0.963	0.976	0.973	0.986	0.991	0.994	0.993	0.994
RGCN	0.933	0.953	0.954	0.970	0.982	0.986	**0.995**	0.994	0.996
HAN	0.908	0.906	0.921	0.957	0.964	0.969	0.988	0.989	0.989
Simple-HGN	0.915	0.960	0.957	0.966	0.980	0.982	0.992	0.992	0.993
HGT	0.910	0.948	0.949	0.959	0.976	0.984	0.994	0.995	0.996
SAGE+AG	**0.973**	**0.985**	0.983	0.972	0.984	**0.995**	0.994	**0.996**	**0.997**
HGT+AG	0.964	0.984	0.990	**0.974**	**0.986**	0.995	0.994	0.994	0.996
Overlap score	0.254	0.937	0.985	0.296	0.922	0.989	0.341	0.932	0.981

6.1 Claim features alone do not reveal rings

When ring claims are drawn like legitimate claims ($\delta = 0$), models that see only the claim's own attributes perform at chance (GB-C 0.084–0.087, MLP-C 0.087–0.095). As δ grows their ring AP rises to about 0.33 ($\delta = 1$) and 0.81 ($\delta = 2$). This confirms that the benchmark separates the two sources of evidence as intended.

6.2 At default ring volume, entity aggregates capture most ring evidence

Adding per-period aggregates of the claim's patient, doctor and hospital (GB-CA) raises ring AP at $\delta = 0$ to 0.889–0.925. This is a consequence of the generator's default ring volume rather than a general property of collusion: ring patients file almost six times as many claims as other patients and see more doctors, and ring doctors file bursts of claims (Section 3.1). Simple counts capture these signals in every regime; when ring volume is reduced, aggregates detect almost nothing (Section 6.6). The remaining gap at default volume is where coordination structure matters.

6.3 Coordination structure helps only when the ring block is dense

Engineered graph features add little at low density but substantially at moderate and high density: at $\delta = 0$, GB-CAG improves on GB-CA by +0.005 ($s = 0.1$, n.s.), +0.078 ($s = 0.5$) and +0.101 ($s = 0.9$; Holm $p = 0.003$). The unsupervised overlap score tells the same story: it is weak at $s = 0.1$ (0.254) but reaches 0.937 and 0.985 at $s = 0.5$ and 0.9 without any training. GNNs show the same pattern: SAGE improves on GB-CA by +0.087 at $\delta = 0$, $s = 0.9$ (Holm $p = 0.003$). Averaged over the

nine regimes, every graph-aware method except HAN outperforms GB-CA (GB-CAG +0.031, SAGE +0.031, RGCN +0.026, Simple-HGN +0.024, HGT +0.021, hybrids +0.039; Holm $p \leq 0.021$ with the seed as unit; HAN +0.007, $p = 0.085$). When ring claims are strongly anomalous ($\delta = 2$), all methods that see claim and aggregate features exceed 0.98 and the differences become negligible.

6.4 GNNs do not outperform gradient boosting with graph features

The central comparison is between message passing and the same structural information supplied as engineered features. Averaged over the regimes, heterogeneous GraphSAGE matches GB-CAG ($\Delta = -0.0001$, 95% CI $[-0.008, 0.008]$), and the other GNNs have lower point estimates (RGCN −0.005, Simple-HGN −0.008, HGT −0.011, HAN −0.024; Table 2). With six seeds none of these differences is significant after Holm adjustment (HAN $p = 0.063$), but every upper CI limit is at most +0.008, so the data are compatible with at most a small GNN advantage. Within single regimes no GNN is significantly better than GB-CAG; SAGE, HAN and Simple-HGN are significantly worse at $\delta = 0$, $s = 0.5$, and SAGE and Simple-HGN at $\delta = 0$, $s = 0.9$ (Holm-adjusted within each nine-regime family).

The hybrids, which add message passing on top of the engineered features, have point estimates of +0.008 over GB-CAG (SAGE+AG 95% CI $[0.002, 0.014]$, HGT+AG $[-0.001, 0.016]$), positive in five of the six seeds, but neither difference is significant after Holm adjustment ($p = 0.12$ and 0.29). The largest difference is in the sparse-structure, no-feature-signal regime ($\delta = 0$, $s = 0.1$: SAGE+AG +0.043, unadjusted $p = 0.011$, Holm $p = 0.095$). Message passing may therefore carry a little information beyond our engineered features, but the evidence is weak, and in this benchmark message passing does not replace them.

Table 2 Paired difference in average precision versus GB-CAG, averaged over the nine regimes: mean $\Delta \pm$ half-width of the 95% t-interval over the six generator seeds, and Holm-adjusted paired t-test p (within each metric). The nine regimes of a seed share its legitimate claims, isolated fraud and ring members, so the seed, not the world, is the unit of replication.

	Ring AP		Isolated AP		Overall AP	
Method	$\Delta \pm$ CI	p	$\Delta \pm$ CI	p	$\Delta \pm$ CI	p
SAGE	0.000 ± 0.008	0.966	−0.085 ± 0.041	0.009	−0.007 ± 0.009	0.299
RGCN	−0.005 ± 0.007	0.364	−0.106 ± 0.044	0.006	−0.013 ± 0.008	0.041
HAN	−0.024 ± 0.015	0.063	−0.147 ± 0.027	<0.001	−0.030 ± 0.012	0.009
Simple-HGN	−0.008 ± 0.008	0.229	−0.094 ± 0.028	0.002	−0.014 ± 0.007	0.022
HGT	−0.011 ± 0.016	0.364	−0.118 ± 0.025	<0.001	−0.018 ± 0.013	0.059
SAGE+AG	+0.008 ± 0.006	0.124	−0.078 ± 0.037	0.009	0.000 ± 0.005	0.935
HGT+AG	+0.008 ± 0.009	0.292	−0.112 ± 0.035	0.002	−0.002 ± 0.006	0.867
MLP-C	−0.560 ± 0.020	<0.001	−0.019 ± 0.024	0.106	−0.402 ± 0.035	<0.001

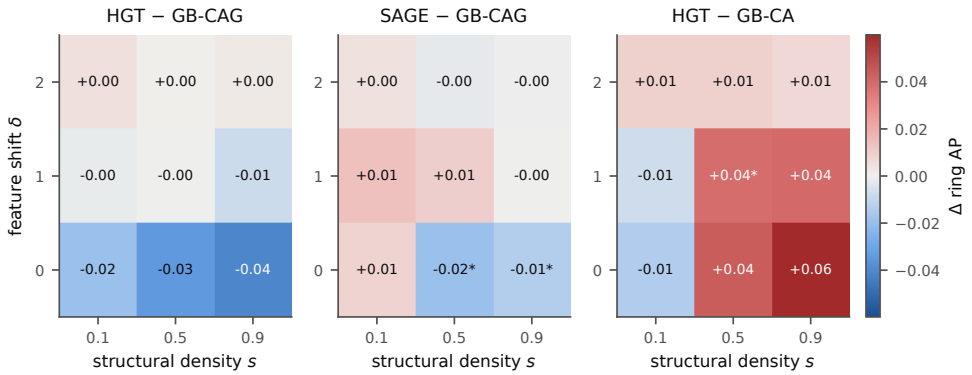


Fig. 4 Mean paired difference in ring AP (six worlds per cell). Left and middle: HGT and SAGE versus gradient boosting with engineered graph features. Right: HGT versus gradient boosting without graph features. *Holm-adjusted $p < 0.05$ within the nine-regime family.

6.5 Message passing dilutes claim-level anomalies

Isolated fraud shows the opposite pattern (Table 3). Every GNN is significantly worse than GB-CAG, by 0.085 (SAGE) to 0.147 (HAN) AP averaged over regimes (Holm $p \leq 0.009$ with the seed as unit), with HAN and HGT the weakest. The MLP control, which sees the claim's own attributes and the static attributes of its patient, doctor and hospital—the GNNs' inputs without degree encodings or message passing—comes within 0.019 of GB-CAG ($p = 0.11$). Aggregating a claim's representation with those of its neighbours, which are overwhelmingly legitimate, blurs the claim's own anomaly; the hybrids, whose claim nodes also see the engineered ratios, recover little of the gap. On all fraud combined (overall AP), GB-CAG has higher AP than every pure GNN (by 0.007–0.030); the difference is significant for RGCN, HAN and Simple-HGN (Holm $p \leq 0.041$) but not for SAGE ($p = 0.30$) or HGT ($p = 0.059$), and GB-CAG does not differ from the hybrids ($p \geq 0.87$).

6.6 When ring members are not unusually active

At default volume, ring members stand out by how many claims they file in every regime (Section 3.1). To test which conclusions depend on this, we repeated the three $\delta = 0$ regimes with ring volume reduced from 16 to 4 claims per month and everything else unchanged. In a test period a ring patient then files 3.7 claims, about the 95th percentile of other patients, a ring doctor 20 claims against 16, and rings make up 2.1% of the non-isolated claims, which is also the chance level of ring AP (Table 4). All 14 methods were run with the hyperparameters of the main experiment and, to check that these did not handicap any method, again after repeating the tuning protocol on low-volume development worlds.

With the hyperparameters of the main experiment, three results change. First, per-entity aggregates no longer detect rings: GB-CA reached a ring AP of 0.049–0.055, barely above chance. Second, coordination structure becomes the main source of evidence for moderately and densely linked rings: GB-CAG reached 0.409 and 0.570

Table 3 Isolated-fraud detection: average precision of isolated-fraud versus legitimate claims (chance level 0.03), mean over six worlds per regime. Bold: best per regime.

	$\delta = 0$			$\delta = 1$			$\delta = 2$		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.471	0.473	0.477	0.560	0.556	0.557	0.644	0.649	0.633
MLP-C	0.542	0.539	0.545	**0.624**	**0.614**	0.623	0.651	**0.667**	0.656
LR-CA	0.375	0.313	0.292	0.556	0.549	0.526	**0.672**	0.665	**0.662**
RF-CA	0.551	0.506	0.469	0.586	0.560	0.547	0.626	0.608	0.604
GB-CA	0.502	0.496	0.477	0.586	0.525	0.540	0.643	0.611	0.627
GB-CAG	**0.575**	**0.613**	**0.628**	0.604	0.605	**0.629**	0.660	0.662	0.655
SAGE	0.447	0.448	0.509	0.536	0.548	0.555	0.615	0.605	0.602
RGCN	0.428	0.425	0.474	0.499	0.516	0.512	0.609	0.602	0.607
HAN	0.375	0.371	0.408	0.472	0.467	0.466	0.590	0.576	0.581
Simple-HGN	0.408	0.401	0.484	0.537	0.518	0.534	0.638	0.639	0.625
HGT	0.344	0.379	0.455	0.500	0.498	0.537	0.622	0.626	0.605
SAGE+AG	0.457	0.503	0.586	0.493	0.548	0.582	0.582	0.584	0.598
HGT+AG	0.378	0.454	0.553	0.444	0.498	0.566	0.549	0.576	0.604
Overlap score	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027

at $s = 0.5$ and 0.9, 0.36 and 0.51 above GB-CA (Holm $p \leq 0.001$), against 0.066 at $s = 0.1$, where no method was far above chance. Third, the pure GNNs recovered little of this structure: at $s \geq 0.5$ their ring AP was 0.058–0.151, only 0.01–0.07 above GB-CA averaged over densities, and 0.23–0.29 below GB-CAG (Holm $p \leq 0.002$ with the seed as unit). One result does not change: no method had significantly higher ring AP than GB-CAG at any density, and the hybrids, which receive the engineered features, matched it (SAGE+AG +0.001, 95% CI $[-0.059, 0.061]$; HGT+AG −0.041 $[-0.098, 0.015]$). On isolated fraud the GNNs were again below GB-CAG, by 0.03–0.08, though not significantly with six seeds.

Re-tuning on low-volume development worlds changed the selected configuration of most methods (HAN, RGCN and Simple-HGN kept theirs and reproduced their results exactly) but not these conclusions (Table 4, right). GB-CAG improved slightly (0.445 and 0.600 at $s = 0.5$ and 0.9, 0.39 and 0.55 above GB-CA), the pure GNNs stayed between 0.058 and 0.162 at $s \geq 0.5$ and 0.26–0.31 below GB-CAG averaged over densities (Holm $p \leq 0.001$), and SAGE+AG matched GB-CAG on average (+0.009 $[-0.048, 0.065]$). The only significant advantage over GB-CAG in either setting was that of SAGE+AG for sparse rings after re-tuning ($s = 0.1$: 0.129 against 0.078, Holm $p = 0.035$), where every method was close to chance. It echoes the default-volume result, where the hybrids' largest gain was also for sparse rings, which suggests that message passing on top of the engineered features helps most where those features are weakest.

6.7 Routing claims to a GNN adds little beyond combining scores

Table 5 and Fig. 5 evaluate two-stage routing (Section 4.4) on the same test claims as the single models. In the design usually proposed, GB-CAG filters and the GNN

Table 4 Ring AP on unseen test rings at $\delta = 0$ when ring volume is reduced from 16 to 4 claims per month (mean over six worlds per cell). At low volume a ring patient files about as many claims as the 95th percentile of other patients and a ring doctor 1.3 times as many as other doctors (default: 5.7 and 2.5 times the averages), and the chance level of ring AP falls from 0.080 to 0.021. Frozen: hyperparameters tuned at default volume; re-tuned: the tuning protocol repeated on low-volume development worlds (hybrids inherit their base configuration). Bold: best per column.

	Default volume			Low volume, frozen			Low volume, re-tuned		
Method / s	0.1	0.5	0.9	0.1	0.5	0.9	0.1	0.5	0.9
GB-C	0.084	0.086	0.087	0.024	0.023	0.022	0.023	0.023	0.022
MLP-C	0.088	0.087	0.095	0.024	0.024	0.024	0.024	0.023	0.024
LR-CA	0.923	0.888	0.868	**0.106**	0.087	0.088	0.090	0.083	0.089
RF-CA	0.875	0.838	0.844	0.057	0.064	0.064	0.062	0.068	0.073
GB-CA	0.925	0.905	0.889	0.049	0.052	0.055	0.047	0.052	0.051
GB-CAG	0.930	0.983	**0.990**	0.066	**0.409**	**0.570**	0.078	**0.445**	**0.600**
SAGE	0.938	0.963	0.976	0.074	0.108	0.131	0.065	0.112	0.162
RGCN	0.933	0.953	0.954	0.066	0.112	0.142	0.066	0.112	0.142
HAN	0.908	0.906	0.921	0.062	0.065	0.058	0.062	0.065	0.058
Simple-HGN	0.915	0.960	0.957	0.074	0.133	0.151	0.074	0.133	0.151
HGT	0.910	0.948	0.949	0.046	0.079	0.098	0.050	0.082	0.108
SAGE+AG	**0.973**	**0.985**	0.983	0.103	0.379	0.566	**0.129**	0.420	0.600
HGT+AG	0.964	0.984	0.990	0.074	0.324	0.523	0.074	0.301	0.477
Overlap score	0.254	0.937	0.985	0.057	0.194	0.281	0.057	0.194	0.281

decides (re-rank). Validation mostly selected small escalation fractions (5% in 30 of 54 worlds for HGT), and the routed system matched GB-CAG (GB-CAG→HGT ΔAP $= -0.001$, 95% CI $[-0.006, 0.003]$; GB-CAG→SAGE $+0.001$). Escalating more claims to HGT lowered both ring AP (from 0.980 at 5% to 0.968 when every claim is escalated) and, more steeply, isolated AP (from 0.626 to 0.508; Fig. 5b–d), because HGT ranks both less well than GB-CAG. Ordering escalated claims by the mean of the two models' percentiles did better: GB-CAG→HGT gained $+0.003$ (Holm $p = 0.60$) and GB-CAG→SAGE $+0.006$ $[0.003, 0.010]$ (Holm $p = 0.042$, positive for all six seeds), on ring ($+0.006$) and isolated fraud ($+0.023$) alike. Validation, however, chose to fuse *every* claim in 22 (HGT) and 30 (SAGE) of the 54 worlds, and on average 55% and 70% of test claims were escalated: the gain comes from combining two different rankings, not from routing a subset of claims. The feature-level hybrids, which give the GNN the engineered features, had slightly higher ring AP ($+0.008$, not significant with the seed as unit) but lost 0.08–0.11 isolated AP and so matched GB-CAG overall. At the ring level all methods reached the ceiling: in all 54 worlds each test ring had at least one claim among the top 5% of claims, and precision in that budget was 0.989–0.998 for every method.

6.8 Cost

Median training time per world (Intel Core i7-1068NG7 laptop CPU, two threads per job with several jobs in parallel, so times are indicative) was 3 s for GB-CAG, 39 s for SAGE, 56 s for RGCN, 86 s for HAN, 126 s for HGT and 144 s for Simple-HGN; scoring

Table 5 Two-stage routing versus single models and feature-level hybrids, averaged over the 54 worlds (same test claims for every row). Routing escalates claims whose GB-CAG score exceeds a threshold set on validation data; the escalation fraction is chosen by validation AP. P@5%: precision among the top 5% of test claims. Esc.: mean share of test claims escalated. ΔAP: paired difference in AP from GB-CAG (± half-width of the 95% t-interval over the six generator seeds), Holm-adjusted p over the rows.

Method	AP	Ring AP	Isol. AP	P@5%	Esc.	ΔAP (p)
GB-CAG	0.947	0.979	0.626	0.996	–	–
HGT	0.929	0.968	0.507	0.989	–	−0.018 ± 0.013 (0.104)
SAGE	0.940	0.979	0.541	0.995	–	−0.007 ± 0.009 (0.597)
HGT+AG	0.945	0.986	0.514	0.996	–	−0.002 ± 0.006 (1.000)
SAGE+AG	0.947	0.987	0.548	0.996	–	0.000 ± 0.005 (1.000)
GB-CAG→HGT, re-rank	0.945	0.979	0.606	0.995	0.16	−0.001 ± 0.005 (1.000)
GB-CAG→HGT, fused	0.950	0.981	0.640	0.997	0.55	+0.003 ± 0.004 (0.597)
GB-CAG→SAGE, re-rank	0.948	0.983	0.595	0.997	0.24	+0.001 ± 0.004 (1.000)
GB-CAG→SAGE, fused	0.953	0.985	0.648	0.998	0.70	+0.006 ± 0.004 (0.042)

a test period took at most 0.25 s for any method. A separate single-thread benchmark on an otherwise idle machine (four worlds at $\delta = 0$) measured scoring directly. For one audit period (about 4,900 claims and 11,200 nodes), building the aggregate and graph features took 0.05 s and the heterogeneous graph 0.01 s, and scoring every claim took 0.05 s with GB-CAG and 0.12 s with HGT. Scoring a single claim, as an online system would, took a median of 2.1 ms (99th percentile 3.1 ms, mostly library-call overhead) with GB-CAG on a precomputed feature row, and 10.7 ms (14.7 ms) with HGT on the claim's exact three-hop computation subgraph (about 126 nodes, 1% of the period graph; extraction 2.8 ms, forward pass 7.7 ms), which reproduces the full-graph score to within 1.2×10^{-7}. Routing would therefore cut the online GNN cost in proportion to the share of claims escalated. In batch scoring the escalated claims' subgraphs overlap: escalating 10% of the claims already touched 55% of the period graph, and scoring that union took 0.08 s against 0.12 s for the whole period. None of these figures includes keeping aggregates, graph features and degree counts up to date as claims arrive, which an online system would need and our code does not implement.

7 Mechanism and Memorisation

7.1 What isolated and ring fraud look like

Fig. 6 shows the test period of one world (seed 1, $\delta = 0$) with the outcomes of GB-CAG and HGT at their validation-selected thresholds. An isolated fraud claim sits on an ordinary patient–doctor edge, so nothing in the graph distinguishes it and only its own attributes can. A ring at $s = 0.1$ links each recruited patient to one or two of the four ring doctors, a pattern that resembles ordinary shared care, whereas at $s = 0.9$ the same members form an almost complete block. The two worlds share the recruited ring members and every legitimate and isolated-fraud claim, and differ only in which ring links exist. In this world GB-CAG flagged 81 and 91 of the ring's 96

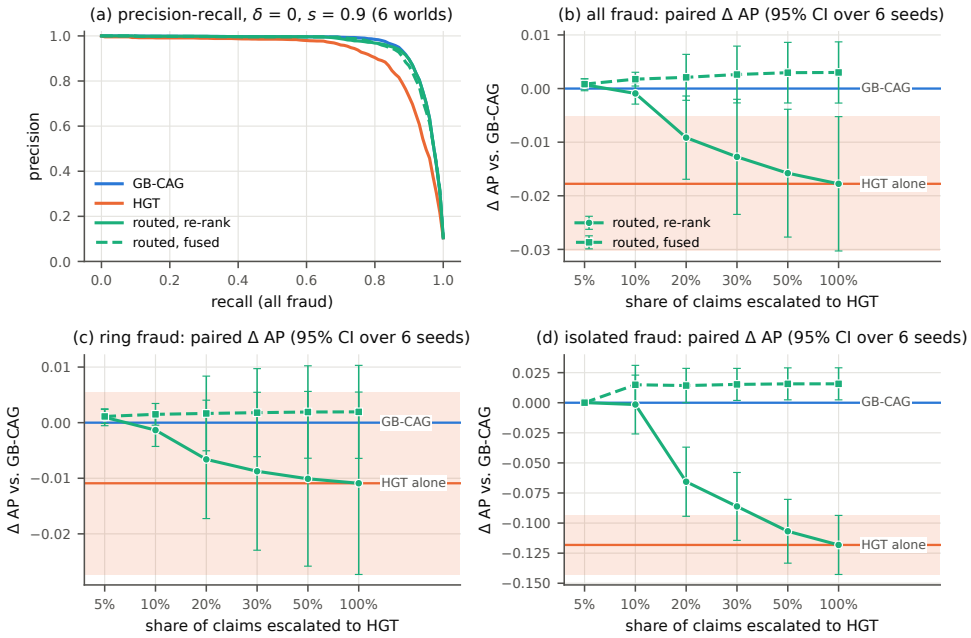


Fig. 5 Two-stage routing from GB-CAG to HGT. (a) Precision–recall curves for all fraud at $\delta = 0$, $s = 0.9$ (precision interpolated at fixed recall and averaged over six worlds; routed systems at the escalation fraction chosen on validation). (b)–(d) Paired difference in test AP from GB-CAG (zero line) against the share of claims escalated, averaged over the 54 worlds, with 95% CIs over the six seeds; orange line and band: HGT alone. The escalation fraction actually used is chosen on validation; the curves over all fractions are shown only to describe the trade-off.

claims at $s = 0.1$ and $s = 0.9$, and HGT 62 and 94: the graph model gains most where the block is dense, as in the averages of Table 1. The figure illustrates one world only; the quantitative statements rest on all 54.

7.2 What the models need

Tables 6 and 7 and Fig. 7 report ablations in the regime where structure matters most ($\delta = 0$, $s = 0.9$), at the default ring volume. Degree-preserving rewiring—permuting which doctor bills each claim within a period, which keeps every patient's and doctor's claim volume but destroys the ring's co-occurrence block—reduces every graph-aware method to the same level (0.757–0.770), close to gradient boosting without graph features after rewiring (0.748). The GNNs therefore do exploit coordination structure, and once it is removed nothing distinguishes them from a model that sees volume alone.

Depth matters: with a single message-passing layer, ring AP falls by 0.135 (SAGE) and 0.161 (HGT); two and three layers perform alike. Removing the direct patient–doctor relation costs little (−0.016 and −0.047, not significant after correction), because two-hop paths through claim nodes carry the same information. The largest effect is specific to HGT: without degree encodings its ring AP drops from 0.949

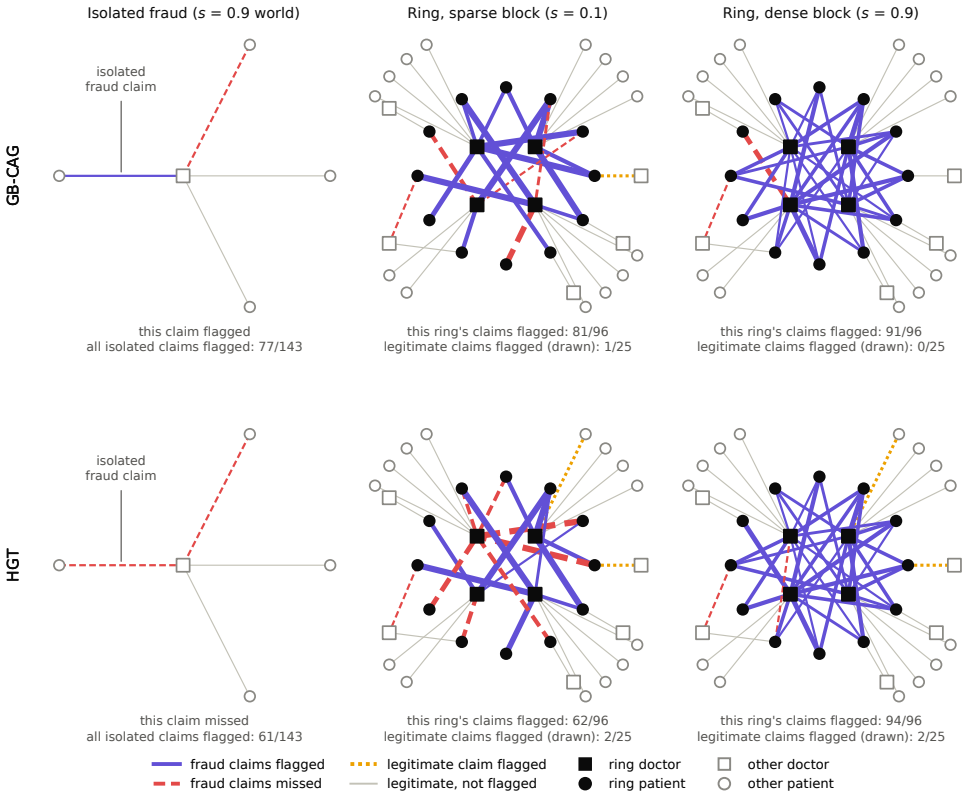


Fig. 6 Isolated and ring fraud in one benchmark world (seed 1, $\delta = 0$; test period), with model outcomes at each model's validation-selected threshold (top: GB-CAG; bottom: HGT). Nodes carry the ground truth (filled: ring member; squares: doctors; circles: patients); edges are patient–doctor pairs with claims in the period, coloured by the outcome for the majority of their claims (line width: number of claims). Left: an isolated fraud claim drawn at random, its patient's doctors and up to ten other patients of its doctor. Middle and right: one test ring, every doctor its patients visit and up to four other patients per ring doctor, at low and high structural density; the two worlds share all recruited members and all legitimate claims. The figure illustrates the generator's typologies; it is not evidence that real rings look alike.

to 0.569 (-0.379, Holm $p = 0.003$), whereas sum-aggregating SAGE changes little (-0.029, n.s.). Softmax attention normalises over neighbours and so cannot count them; supplying counts as input features, or aggregating by summation, restores the signal [38].

7.3 Which parts of HGT matter

Table 7 changes one factor of the tuned HGT at a time in the same regime. Depth beyond two layers and the number of attention heads made no significant difference (four layers $+0.002$; one, two and eight heads -0.037 to $+0.013$; all Holm $p > 0.5$), so there is no sign of over-smoothing up to four layers on these graphs. Among the relations, removing the patient–claim edges cost most (-0.219, Holm $p = 0.005$);

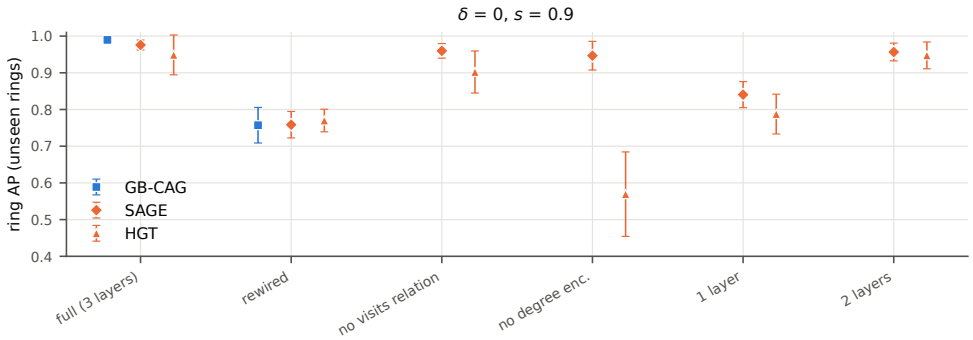


Fig. 7 Mechanism ablations at $\delta = 0$, $s = 0.9$ (mean and 95% t-interval over six worlds). Rewiring preserves every node's degree but destroys the ring block.

Table 6 Mechanism ablations at $\delta = 0$, $s = 0.9$ (six worlds): ring AP (mean ± s.d.) and paired change from the full model, with Holm-adjusted p over every ablation comparison in this table and Table 7. The tuned GNNs use three layers; HGT's single-factor variants are in Table 7.

Method	Variant	Ring AP	Δ vs. full
GB-CA	Full model	0.889 ± 0.026	–
	Degree-preserving rewiring	0.748 ± 0.027	−0.141 ($p = 0.018$)
	+ ring identifier (legacy leak)	1.000 ± 0.000	+0.111 ($p = 0.003$)
GB-CAG	Full model	0.990 ± 0.008	–
	Degree-preserving rewiring	0.757 ± 0.046	−0.232 ($p = 0.002$)
SAGE	Full model	0.976 ± 0.013	–
	Degree-preserving rewiring	0.759 ± 0.034	−0.217 ($p = 0.002$)
	No patient–doctor relation	0.960 ± 0.019	−0.016 ($p = 0.360$)
	No degree encodings	0.947 ± 0.037	−0.029 ($p = 1.000$)
	1 message-passing layer	0.841 ± 0.034	−0.135 ($p = 0.002$)
	2 layers	0.957 ± 0.023	−0.019 ($p = 0.413$)
HGT	Full model	0.949 ± 0.052	–
	Degree-preserving rewiring	0.770 ± 0.029	−0.179 ($p = 0.036$)
	No reverse relations (legacy graph)	0.772 ± 0.024	−0.177 ($p = 0.022$)

removing the claim–doctor edges cost 0.113 on rings (Holm $p = 0.085$) and 0.147 on isolated fraud ($p < 0.001$), while hospital nodes and the direct patient–doctor relation contributed nothing measurable. The node inputs separate the two tasks cleanly. With claim attributes set to zero, ring AP was unchanged (−0.012) and isolated AP fell to chance (0.036); an HGT that saw only the degree encodings and the graph reached a ring AP of 0.982 ± 0.018, as high as the full model, and detected no isolated fraud. HGT's ring detection is therefore structural—counts of who bills whom, propagated along patient–claim–doctor paths—and its claim attributes serve isolated fraud only. This is the information that the engineered shared-patient features summarise directly. At default volume these counts also carry the ring members' excess claims; Section 6.6

shows that when they do not, the GNNs fall far behind gradient boosting with the engineered features. HGT also varied between worlds (standard deviations up to 0.11), so with six worlds differences smaller than about 0.05 cannot be resolved.

Table 7 HGT single-factor ablations at $\delta = 0$, $s = 0.9$ (six worlds; one factor changed at a time from the tuned configuration: 3 layers, 4 heads, hidden size 32). Ring AP (mean ± s.d.), isolated AP (mean) and paired change from the full model; Holm-adjusted p over every ablation comparison in this table and Table 6.

Variant	Ring AP	Δ ring AP	Isol. AP	Δ isol. AP
Full model	0.949 ± 0.052	–	0.455	–
Depth (full: 3)				
1 layer	0.788 ± 0.052	−0.161 ($p = 0.020$)	0.239	−0.216 ($p = 0.007$)
2 layers	0.947 ± 0.035	−0.001 ($p = 1.000$)	0.423	−0.032 ($p = 1.000$)
4 layers	0.951 ± 0.043	+0.002 ($p = 1.000$)	0.461	+0.005 ($p = 1.000$)
Heads (full: 4)				
1 attention head	0.920 ± 0.049	−0.028 ($p = 1.000$)	0.384	−0.071 ($p = 0.228$)
2 heads	0.911 ± 0.081	−0.037 ($p = 0.506$)	0.473	+0.018 ($p = 0.813$)
8 heads	0.961 ± 0.023	+0.013 ($p = 1.000$)	0.524	+0.069 ($p = 0.282$)
Relations				
No patient–claim relation	0.730 ± 0.055	−0.219 ($p = 0.005$)	0.311	−0.144 ($p = 0.333$)
No claim–doctor relation	0.836 ± 0.037	−0.113 ($p = 0.085$)	0.308	−0.147 ($p < 0.001$)
No patient–doctor relation	0.902 ± 0.055	−0.047 ($p = 0.413$)	0.423	−0.032 ($p = 1.000$)
No hospital nodes	0.946 ± 0.032	−0.002 ($p = 1.000$)	0.427	−0.028 ($p = 1.000$)
Node inputs				
No degree encodings	0.569 ± 0.110	−0.379 ($p = 0.003$)	0.375	−0.080 ($p = 0.813$)
Claim attributes zeroed	0.937 ± 0.064	−0.012 ($p = 1.000$)	0.036	−0.419 ($p < 0.001$)
Entity attributes zeroed	0.969 ± 0.027	+0.020 ($p = 1.000$)	0.444	−0.011 ($p = 1.000$)
Degree encodings only	0.982 ± 0.018	+0.033 ($p = 1.000$)	0.040	−0.415 ($p < 0.001$)

7.4 Which inputs the trained models use

To see which inputs the trained models use, without retraining, we permuted one input group at a time across the test period, five times each, and recorded the drop in ring AP: for GB-CAG the C, A and G features and each G feature separately, and for the GNNs the claim attributes, static entity attributes and degree encodings across nodes of each type. At $\delta = 0$, GB-CAG relied on aggregates when rings were sparse (A: −0.80 at $s = 0.1$; G: −0.13) and switched to the graph features as the block became dense (G: −0.81 and −0.88 at $s = 0.5$ and 0.9; A: below 0.06), above all the normalised triadic closure of the claim (−0.47 on its own at $s = 0.9$). SAGE and HGT relied almost entirely on degree encodings (−0.76 to −0.89 at every density) and hardly at all on claim or entity attributes (below 0.05). The two kinds of score are not on a common scale: GB-CAG's measure reliance on engineered summaries of the graph, whereas a GNN's degree encodings are raw counts whose relational meaning is extracted by message passing, and permutation understates groups whose information is duplicated elsewhere. Read together with the ablations, they are consistent with both model

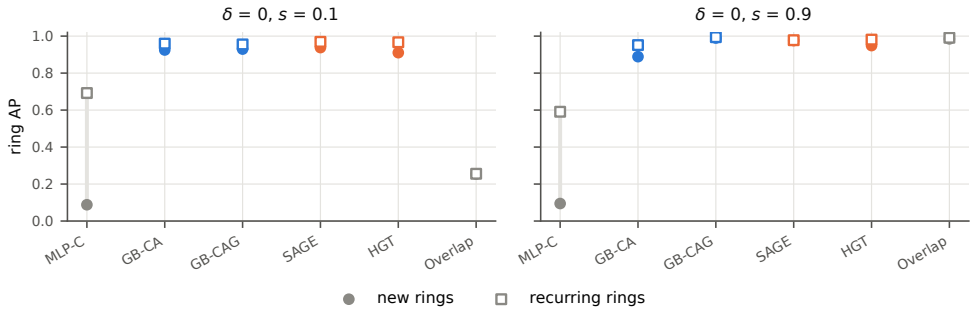


Fig. 8 Ring AP on new versus recurring rings ($\delta = 0$, six worlds).

families detecting dense rings from the same evidence: how many counterparties a claim's patient and doctor have, and how many of them they share. (Full results, including isolated fraud and test-time rewiring, are in the repository.)

7.5 Two evaluation pitfalls, quantified

The mechanism ablations (Table 6) also quantify two defects that are easy to introduce. Adding the generator's ring identifier to the feature set—a column that is non-zero only for ring claims—raises gradient boosting from 0.889 to a perfect 1.000. Building the heterogeneous graph with one-directional relations only (patient→claim→doctor→hospital) leaves claim nodes reachable only from patients, so doctor context never reaches them; HGT then loses 0.177 AP. Both defects produce plausible-looking numbers and neither is caught by aggregate metrics alone; the tests listed in Section 10 detect both.

7.6 Recognising known members is not detecting rings

Table 8 and Fig. 8 compare new rings (emergent protocol) with rings whose members recur from training (persistent protocol). The MLP control, which sees only the claim and the static attributes of its patient, doctor and hospital, rises from chance to 0.692 ($s = 0.1$) and 0.591 ($s = 0.9$) when rings recur: it identifies the members from their attributes. HGT gains 0.056 at $s = 0.1$ (95% CI $[0.029, 0.084]$), and gradient boosting with aggregates gains 0.035–0.063, while the unsupervised overlap score, which cannot memorise, is unchanged. Evaluations in which test rings share members with training rings therefore overstate how well a model detects coordination, by an amount that differs between methods and can reorder them.

8 Results on the Public Provider-Labelled Data

Table 9 and Fig. 9 report provider-level results over the five rotations (1,082 test providers each, about 101 fraudulent). Gradient boosting on provider aggregates (GB-P) has the highest mean AP (0.704 ± 0.045). Adding label-free graph features (GB-PG, -0.009, corrected 95% CI $[-0.042, 0.023]$) or graph and label-neighbourhood

Table 8 Ring AP when test rings are new (emergent) versus recurring members of training rings (persistent), $\delta = 0$, six worlds each; gain = persistent − emergent (paired over worlds, 95% t-interval). The overlap score is unsupervised and cannot memorise.

	$s = 0.1$			$s = 0.9$		
Method	new	recurring	gain	new	recurring	gain
MLP-C	0.088	0.692	+0.604 [+0.523, +0.685]	0.095	0.591	+0.496 [+0.416, +0.577]
GB-CA	0.925	0.959	+0.035 [+0.014, +0.055]	0.889	0.952	+0.063 [+0.034, +0.092]
GB-CAG	0.930	0.955	+0.025 [−0.012, +0.062]	0.990	0.994	+0.005 [−0.002, +0.012]
SAGE	0.938	0.969	+0.031 [−0.019, +0.081]	0.976	0.978	+0.002 [−0.015, +0.020]
HGT	0.910	0.967	+0.056 [+0.029, +0.084]	0.949	0.981	+0.032 [−0.017, +0.082]
Overlap score	0.254	0.256	+0.002 [−0.137, +0.142]	0.985	0.991	+0.006 [−0.009, +0.020]

features (GB-PGL, −0.007 [−0.030, 0.016]) does not improve it. Every GNN has lower AP than GB-P, by 0.029 (HGT) to 0.083 (RGCN), and the hybrids, which give the GNN the provider and graph features, are no better than the plain GNNs. With five overlapping rotations, none of the 24 differences is significant after the correction and Holm adjustment (smallest Holm $p = 0.13$, SAGE+G versus GB-P); a plain paired t-test, which ignores the overlap, would have called SAGE+G significantly worse (Holm $p = 0.021$). Holding the trained models fixed, the provider-level bootstrap places the AP of every GNN and hybrid, and of GB-PG, below that of GB-P (95% intervals from $[-0.108, -0.055]$ for RGCN to $[-0.048, -0.006]$ for HGT), so the shortfall does not depend on which providers happened to be tested; whether it would survive retraining cannot be resolved with five rotations. ROC-AUC lies between 0.927 and 0.942 for all methods and recall among the top 10% of providers between 0.59 and 0.64, so the methods rank providers similarly; the AP differences come from the top of the ranking. Under the same parallel load as before, median training time per rotation was 2–15 minutes for the GNNs (range 1.4–24 minutes) against 7 s for GB-P. Because the tuning split shares providers with the test folds, we re-selected the configuration of every tabular and MLP model inside each rotation, using only that rotation's validation fold. AP changed by at most 0.010 (GB-P 0.706 versus 0.704; no paired difference significant), and GB-P remained the best method, so the overlap does not explain the ranking. The GNNs were not re-selected per rotation, which would take about 25 CPU-hours; because selecting on providers that later appear in test folds is expected to make the selected configuration look better on them, the shared selection favours the GNNs, and their shortfall is, if anything, understated.

Table 9 Provider-level fraud detection on the public provider-labelled data (5-fold rotation; mean ± s.d. over folds). F1 at the validation-selected threshold; Recall@10%: share of all fraudulent providers that rank among the top 10% of providers by score.

Method	AP	ROC-AUC	F1	Recall@10%
LR-P	0.678 ± 0.041	0.940 ± 0.017	0.617 ± 0.030	0.644 ± 0.023
RF-P	0.681 ± 0.049	0.942 ± 0.017	0.624 ± 0.031	0.644 ± 0.025
GB-P	0.704 ± 0.045	0.939 ± 0.020	0.624 ± 0.028	0.636 ± 0.027
GB-PG	0.695 ± 0.040	0.937 ± 0.019	0.600 ± 0.043	0.642 ± 0.024
GB-PGL	0.697 ± 0.042	0.937 ± 0.019	0.595 ± 0.045	0.634 ± 0.019
MLP-P	0.685 ± 0.052	0.940 ± 0.019	0.613 ± 0.028	0.642 ± 0.032
SAGE	0.664 ± 0.047	0.938 ± 0.018	0.605 ± 0.036	0.624 ± 0.046
RGCN	0.621 ± 0.054	0.927 ± 0.019	0.578 ± 0.029	0.593 ± 0.029
HAN	0.666 ± 0.046	0.938 ± 0.018	0.583 ± 0.028	0.636 ± 0.045
Simple-HGN	0.670 ± 0.057	0.939 ± 0.020	0.600 ± 0.037	0.630 ± 0.036
HGT	0.676 ± 0.058	0.939 ± 0.019	0.613 ± 0.051	0.640 ± 0.051
SAGE+G	0.653 ± 0.048	0.937 ± 0.018	0.612 ± 0.040	0.623 ± 0.031
HGT+G	0.648 ± 0.032	0.933 ± 0.014	0.589 ± 0.022	0.615 ± 0.035

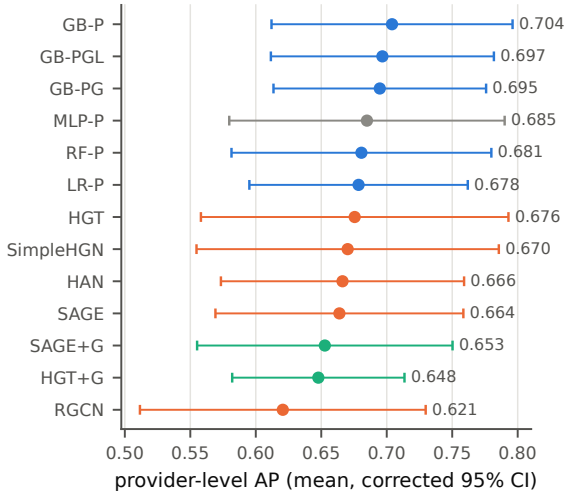


Fig. 9 Provider-level AP on the public provider-labelled data (mean over five rotations; 95% CI from the corrected resampled t-distribution, which accounts for the overlap of the rotations' training sets). Blue: tabular; orange: GNN; aqua: hybrid; grey: MLP.

The graph statistics explain why. Almost every provider (99.5%) shares beneficiaries with other providers (median 30 co-providers), but labels are not assortative through this relation: on average 23.9% of a fraudulent provider's co-providers are labelled fraudulent, against 25.3% for a legitimate provider (both well above the 9.4% base rate because shared beneficiaries concentrate on large providers). Shared physicians show mild assortativity (19.1% versus 14.2%) but connect only 73% of providers, with a median of two co-providers. Fraudulent providers are, however, much larger (median 156 versus 27 claims), a volume signal that provider aggregates capture

directly. In the terms of our benchmark, these data resemble the low-density regime, where structure added nothing beyond per-entity aggregates. The labels mark potentially fraudulent providers without identifying rings, so the absence of a graph benefit here does not show that rings are absent from Medicare data; it shows that this public labelling contains no exploitable relational signal for the models tested.

9 Discussion

9.1 When does relational structure help?

Three conditions emerge from the controlled study. First, the trace of coordination must be dense enough to distinguish a ring from ordinary shared care: graph information added 0.04–0.10 AP at moderate and high density at default volume (most when ring claims looked legitimate) and 0.36–0.55 when ring volume was low, but nothing measurable at low density. Second, the value of structure depends on what else distinguishes ring members: at default volume their elevated utilisation, which per-entity counts capture, already identified most rings, whereas at low volume aggregates were close to chance and coordination structure carried almost all of the evidence. Third, the claims must not already be anomalous on their own: when ring claims were strongly upcoded ($\delta = 2$), every model with claim and aggregate features exceeded 0.98 AP and structure became redundant.

9.2 Why GNNs did not beat engineered graph features

When structure did help, gradient boosting with a handful of shared-patient statistics captured it as well as or better than the best GNN. The mechanism analyses suggest why. The useful signal here is largely *counting*—how many of a patient's doctors also share other patients—and attention- or mean-based aggregators cannot count: HGT lost most of its ring detection without degree encodings, whereas sum-aggregating GraphSAGE barely changed, and an HGT given only degree encodings and the graph detected rings as well as the full model (Section 7). This mirrors the expressiveness argument for sum aggregation [38]. GNNs also dilute claim-level anomalies through neighbourhood averaging, which made them markedly worse on isolated fraud, and they readily memorise the attributes of training-ring members, which inflates performance when rings recur but does not transfer to new rings. At default volume, hybrids that combine message passing with engineered features had a small gain over gradient boosting on ring fraud that was not significant with the seed as unit; whatever structure message passing extracts beyond our features is small, and it comes at 9–40 times the training cost.

The low-volume stress test sharpens this picture. Once ring members no longer stood out by volume, counting no longer sufficed, and the pure GNNs recovered only a small part of the co-occurrence structure that the engineered closure features summarise, with the main hyperparameters and after re-tuning alike, whereas hybrids given those features came close to or matched gradient boosting and, for sparse rings, exceeded it. Message passing over two training periods with about 200 ring claims did not learn the shared-patient pattern that a single engineered feature states directly:

at $s \geq 0.5$ the unsupervised overlap score alone (0.194 and 0.281) had higher ring AP than every pure GNN in both settings.

One caveat favours the engineered features: we designed them knowing the injected typology, a dense doctor–patient block. Their success shows that a few targeted features suffice when the typology is known; it does not show that they would capture typologies they were not designed for, where learned message passing could have an advantage. These findings are consistent with evidence from other domains that tree ensembles with neighbourhood features are strong graph anomaly detectors [10, 11, 32], with the same pattern on public Medicare provider data [12], and extend it to heterogeneous patient–provider claim graphs with controlled typologies.

9.3 Why a small fixed benchmark can point the other way

To show how a small, fixed benchmark can mislead, we re-analysed one: a synthetic dataset of 2,100 claims with five injected rings, produced by an earlier, simpler generator of ours that is included in the repository, evaluated with its own date-sorted 60/20/20 split (Table 10). Such a dataset appears to test ring detection, but this one cannot tell whether relational structure helps, for three reasons. First, its ring claims were drawn from the same inflated distribution as its isolated fraud, so they were anomalous on their own: claim amount alone separates ring from legitimate claims with ROC-AUC 0.94, more than in our most anomalous regime ($\delta = 2$, 0.90), and gradient boosting on three raw claim attributes ranks all 21 test ring claims above every legitimate claim (ring AP 1.000). Second, the same five rings were active for all 24 months, so every test ring claim was billed by a doctor with labelled ring claims in training, which is the memorisation setting of Section 7. Third, the test split held 21 ring claims (3–6 per ring): one claim moves ring recall by 4.8 points, even a perfect recall has an exact 95% interval of $[0.84, 1.00]$, so a recall gap of a few claims between two models is within noise, and a single dataset says nothing about variation between datasets. None of these properties is peculiar to that dataset. A small, fixed benchmark whose injected rings differ from legitimate claims in their features, recur across the split and contribute a few dozen test claims cannot separate relational learning from claim-level anomaly detection or from recognising known members. Controlling the feature-level signal, testing on new rings and replicating worlds removes all three confounds. Because the generator, features, evaluation unit and tuning differ, numbers from the two set-ups are not directly comparable.

9.4 Public provider-labelled data

On the public provider-labelled data, gradient boosting on provider aggregates had the highest AP, and neither engineered graph features, label-neighbourhood features nor any GNN improved on it, in agreement with earlier comparisons on these data [12, 30]. The labels are not assortative through shared beneficiaries, and fraudulent providers differ mainly in volume, so the data sit in the regime where our benchmark also found no benefit from structure. This is consistent with the controlled results but is not an independent confirmation of the dense-ring regime: validating that regime requires real data with labelled collusion, which are not publicly available. Because the source

Table 10 A small fixed benchmark (2,100 claims, five rings; an earlier, simpler generator of ours, included in the repository), re-analysed with its own split (date-sorted, 60/20/20), compared with our benchmark. Benchmark ranges span $\delta = 0$ to $\delta = 2$ ($s = 0.5$, six worlds each; ring AP and recall of GB-C pooled over s). Shift and ROC-AUC use raw claim amounts, not adjusted for specialty, for both. Interval: exact 95% binomial.

	Fixed dataset	This benchmark
Claims in total (in the test split / period)	2,100 (420)	20,176 (4,924), mean per world
Ring claims in test	21 (3–6 per ring)	384 (4 rings × 96)
Rings	5, active all 24 months	4 per period, each active 6 months
Test-ring doctors with labelled ring claims in training	100%	0% (by design)
Ring vs. legitimate log-amount shift (s.d.)	1.49	0.11 ($\delta = 0$) to 1.87 ($\delta = 2$)
ROC-AUC of claim amount alone, ring vs. legitimate	0.94	0.53 to 0.90
Ring AP, GB on claim attributes only	1.000	0.086 to 0.811
Ring recall, GB on claim attributes only	21/21 [0.84, 1.00]	0.15 to 0.76
Independent datasets	1	54

of the claims and of the labels is undocumented, the result should also not be read as an estimate of how these methods perform on current Medicare claims.

9.5 Practical recommendations

- Start with gradient boosting on claim features, per-entity aggregates and a small set of shared-patient graph features; on our benchmark no pure GNN was significantly better in any regime, at default or low ring volume, the hybrids gained significantly only for sparse rings at low volume, and gradient boosting was 9–50 times cheaper to train than the GNNs and hybrids. Where rings may be sparse, a hybrid may add a little, although every method detected sparse rings poorly when they were not unusually active.
- Do not infer the value of relational structure from rings whose members are unusually active: then aggregates already find them. When they are not, explicit co-occurrence features, or hybrids that receive them, were needed.
- Evaluate on rings (or providers) that do not appear in training, and compare against a no-graph control with identical inputs; otherwise memorisation of known members can masquerade as relational learning. Report uncertainty over the independent units—generated worlds that share data, or cross-validation rotations that share training data, are not independent.
- If a GNN is used, include reverse relations, verify that information reaches the classified nodes, and give attention-based architectures structural (degree) encodings or prefer sum aggregation.
- Keep isolated-fraud detection on a claim-level model; message passing degraded it consistently.

- Do not expect a GNN second stage to pay for itself. Letting a GNN re-rank the claims that gradient boosting flags gained nothing here; averaging the two models' rankings gained at most 0.006 AP, and in an audit setting, where a whole period is scored at once, routing saves little computation (Section 6).

For a payer's health information system these results favour a modest design. The aggregates and shared-patient statistics used here are counts and overlaps over one audit period of claims, which a claims database can compute in batch, and the resulting model scored a whole period in a fraction of a second on a laptop CPU (Section 6). A GNN adds a graph-learning stack and 13–50 times the training cost of gradient boosting in our runs; a pure GNN brought no measurable gain in our benchmark or on the public data, and a hybrid only for sparse, low-volume rings. Because investigators examine only the highest-ranked cases, such systems should be evaluated at the investigation budget and on providers and rings unseen in training.

9.6 Limitations

The controlled findings hold for our generator. Its distributions are simple, labels are noise-free and immediate, rings are static during their active period and of a single size, and ring prevalence (8% of the non-isolated test-period claims at default volume) is far above realistic levels so that each world has enough positives for stable estimates. At default volume ring members carry a strong volume signal in every regime, so the nine regimes vary the *additional* structural evidence rather than all of it; the low-volume stress test (2.1% prevalence) covers a single reduced volume and only $\delta = 0$. The engineered graph features were designed with knowledge of the injected typology, which favours them over learned message passing (Section 9). We tuned each method once, on development worlds from a single regime ($\delta = 0.5$, $s = 0.5$), and once more for the low-volume test; per-regime tuning could change small differences. The six seeds per regime are the unit of replication: they give limited power within a regime, and because the regimes of a seed share most of their data, pooled tests have five degrees of freedom and cannot resolve differences much smaller than 0.01–0.02 AP. Each model was trained once per world. Graph models were trained full-batch on graphs of about 11,000 nodes per period; larger graphs would need sampling-based training [40]. The architecture and reliance analyses were run in one regime ($\delta = 0$, $s = 0.9$ for the ablations) with one model seed per world, and permutation importance can understate a feature group whose information is duplicated in another; these analyses describe what the trained models use, not causal effects of the generator. Routing was evaluated with one first stage, two GNNs and a grid of six escalation fractions; the GNNs' validation scores come from the period used for early stopping, which slightly favours escalation. Cost was measured on a single laptop CPU thread and excludes the upkeep of aggregates, graph features and degree counts as claims arrive, data retrieval and concurrency, none of which exist in our code. On the public data, labels are per provider, the source of the claims and of the labels is undocumented, the observation window is a single year, and five overlapping rotations give the corrected tests little power. Hyperparameters were selected on a split whose validation providers overlap the evaluation test folds; re-selecting the tabular and MLP configurations inside each

rotation (Section 8) removes this overlap for those models, but the GNNs keep the shared selection, which if anything favours them. Finally, the benchmark does not model adaptive fraudsters, temporal ring dynamics, which temporal graph models target [49], or realistic coding semantics such as those of synthetic health records [50].

10 Reproducibility

All results are produced by one released package from raw inputs. Every run writes a record containing the git commit, a hash of its configuration, the hashes of the generated claim and ground-truth tables, the library versions and the per-claim (or per-provider) test predictions; the tables and figures of this paper are generated from these records by a single script. Automated tests, run in continuous integration, check that (i) no ground-truth column reaches any feature matrix and features are invariant to permuting the labels; (ii) a period's features and graph are unchanged when claims of other periods are perturbed; (iii) emergent rings in different splits share no doctor or patient; (iv) at $\delta = 0$ ring claims are distributed like legitimate claims; (v) features, graphs and labels stay aligned by claim identifier; and (vi) for every GNN, perturbing doctor features changes claim logits, so relational information actually reaches the classified nodes; (vii) a routed score depends only on the claim's own two scores and validation statistics; (viii) rewiring the test period leaves the features of other periods unchanged; (ix) scoring a claim on its L-hop computation subgraph reproduces its full-graph score; and (x) the regimes of a seed share their legitimate claims and ring members, the design that makes the seed the unit of pooled tests. A further check confirms that every table in this article equals the file generated from the run records. The original runs were made from a working tree that had not yet been committed, so their records do not pin the code version; retraining GB-CAG, SAGE and HGT with the released code on all 54 worlds reproduced their stored test predictions (Section 4.4), which ties the reported numbers to the released code. Records now also store a hash of the research sources at process start. A checklist mapping this study to the REFORMS recommendations [51] and a datasheet for the generator [52] are included in the repository.

11 Conclusion

We asked when relational structure helps healthcare fraud detection and answered it with a benchmark that varies the feature-level and structural signal of provider rings independently, tests on rings never seen in training, and checks the conclusions on public provider-labelled data. Structure helped only when rings formed moderately or densely linked doctor–patient blocks, and it mattered most when ring members were not otherwise unusual: then per-entity aggregates detected almost nothing. In every setting a gradient-boosting model with a handful of engineered shared-patient features matched or outperformed five heterogeneous GNNs at a fraction of the cost, and at low ring volume it detected far more rings than any pure GNN; adding message passing on top of those features, or routing high-risk claims to a GNN, gained little. GNNs generalised to new rings only with reverse relations, sufficient depth and counting ability, degraded isolated-fraud detection, and could memorise known ring members

in ways that inflate conventional evaluations. A small fixed benchmark whose rings are anomalous in their features, recur across the split and contribute few test claims can suggest the opposite conclusion. On public provider-labelled data no graph-based method improved on tabular aggregates, as earlier work on these data also found. These conclusions rest on a simulator whose engineered features were designed for its typology. We recommend that graph models for claims fraud be justified against a graph-augmented tabular baseline and a no-graph control, on entities unseen in training, with uncertainty computed over independent units. The benchmark, code and tests are publicly available to support such comparisons; the most valuable next step is real data with labelled collusive schemes.

Supplementary information. The repository contains the per-world results (including standard deviations, F1, ROC-AUC and recall at a fixed investigation budget for every regime and method), the tuning logs and the run records from which every table and figure is generated. The per-claim and per-provider test predictions (about 30 MB) are not tracked in the repository; the released code regenerates them.

Statements and Declarations

- **Funding.** The authors did not receive support from any organization for the submitted work.

- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable: the study uses synthetic data and a public dataset released under a CC0 licence, in which beneficiaries, providers and physicians are identified only by pseudonymous codes; no human participants were involved and no new data were collected.
- **Consent for publication.** Not applicable: the manuscript contains no individual person's data in any form; it uses synthetic data and a public dataset with pseudonymous identifiers only.
- **Data availability.** The synthetic benchmark, including the low-volume setting, is generated by the released code from fixed seeds. The public provider-labelled data are available from Kaggle [47] under CC0; file checksums are listed in the repository.
- **Code availability.** Code, configurations, tests and result records are available at https://github.com/akshay-hudev/Fraud-Guard (directory `research/`) under the MIT licence. An archived release with a DOI will be provided upon acceptance.
- **Author contribution.** H M Akshay: conceptualization, methodology, software, investigation, formal analysis, data curation, visualization and writing—original draft. Gaurav S B, Eshita Narayana and Harshita Srivastava: methodology, investigation, formal analysis and writing—original draft. Malini M Patil: supervision and writing—review and editing. All authors read and approved the final manuscript.

Appendix A Generator parameters and hyperparameters

Table A1 Generator parameters (defaults in `research/synth/generator.py`).

Parameter	Value
Patients / doctors / hospitals / regions / specialties	6,000 / 300 / 40 / 5 / 8
Audit periods	4 × 6 months (train: 1–2, validation: 3, test: 4)
Legitimate claims per patient	Poisson(gamma rate; mean 3, shape 1.5) over 24 months
Share of legitimate claims with a regular doctor	0.85 (1–3 regular doctors per patient, same region)
Claim amount	log-normal: specialty mean + 0.15 (procedures − 1) + 0.8 inpatient, s.d. 0.6
Isolated fraud	3% of legitimate volume; feature shift 2.0
Ring size	4 doctors, 12 recruited patients (utilisation-weighted)
Ring volume	16 claims per month during activity; monthly burst days (low-volume stress test: 4)
Rings per period	4 (emergent: new members each period; persistent: same 4 rings)
Feature shift δ	0, 1, 2 (development worlds: 0.5)
Structural density s	0.1, 0.5, 0.9 (development worlds: 0.5)
Worlds per regime	6 (seeds 1–6; the regimes of a seed share its legitimate claims, isolated fraud and ring members); development worlds: seeds 901, 902

Table A2 Selected hyperparameters (equal budget per method; hybrids inherit their base configuration; methods sharing a search space were offered the same candidate configurations). Synthetic: tuned on development worlds; public data: tuned on a separate split. The configurations re-tuned for the low-volume stress test are in the repository (`results/tuning_lowvolume`).

Method	Synthetic benchmark	Public data
LR	C=100	C=1
RF	max depth=16; max features=sqrt; min samples leaf=3; n estimators=300	max depth=16; max features=0.3; min samples leaf=10; n estimators=300
GB (C)	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=10	–
GB (C+A / P)	l2=1; learning rate=0.1; max iter=200; max leaf nodes=63; min samples leaf=50	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (C+A+G / P+G)	l2=1; learning rate=0.1; max iter=400; max leaf nodes=15; min samples leaf=20	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
GB (P+G+L)	–	l2=0; learning rate=0.03; max iter=200; max leaf nodes=15; min samples leaf=50
MLP	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05	dropout=0.3; hidden=128; layers=2; lr=0.001; weight decay=1e-05
SAGE	aggr=sum; dropout=0.1; hidden=32; layers=3; lr=0.003; weight decay=0.001	aggr=mean; dropout=0.3; hidden=64; layers=3; lr=0.003; weight decay=0.0001
RGCN	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.003; weight decay=0.001	aggr=sum; dropout=0.3; hidden=64; layers=2; lr=0.01; weight decay=0.001
HAN	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=2; hidden=64; layers=2; lr=0.003; weight decay=0.0001
Simple-HGN	dropout=0.3; heads=4; hidden=64; layers=3; lr=0.003; weight decay=0.0001	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001
HGT	dropout=0.3; heads=4; hidden=32; layers=3; lr=0.01; weight decay=1e-05	dropout=0.1; heads=4; hidden=64; layers=2; lr=0.01; weight decay=0.001

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